

PhD Proposal

Developing an Explainable Deep Learning Framework for Forecasting the Implications of Digital Transformation on Employment Dynamics, Productivity, and Workforce Skills in Zimbabwe's Banking Sector

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1. Raw / Input Features

These are features collected directly from bank systems, HR systems, surveys, reports, AI systems, and operational records before transformation.

1.1 Productivity Raw Features

1. Total digital revenue
2. Number of digital transactions
3. Number of manual transactions
4. Total new bank accounts
5. Total customer queries
6. Total mobile banking transactions
7. Total internet banking transactions
8. Total agent banking transactions
9. Total card transactions
10. Total POS transactions
11. Total loan applications received
12. Total loans approved
13. Total loans declined
14. Total loan disbursements
15. Number of active digital customers
16. Number of call centre interactions
17. Number of complaints received
18. Number of complaints resolved
19. Average service turnaround time
20. Number of failed transactions
21. Number of reversed transactions
22. Number of fraud cases detected
23. Number of compliance cases processed
24. Number of KYC cases processed
25. Total revenue
26. Total operating cost
27. Total income
28. Non-interest income
29. Fee and commission income
30. Loan book
31. Number of loans
32. Total deposits
33. Total assets
34. Total liabilities
35. Total branch transactions
36. Total ATM transactions
37. Total inward transfers
38. Total outward transfers
39. Total loan repayments received
40. Total withdrawals
41. Total cash deposits
42. Total electronic deposits
43. Number of active customers
44. Number of inactive customers
45. Number of branch customers served
46. Number of accounts closed

47. Number of dormant accounts reactivated
48. Total bank accounts
49. Digital active user growth records
50. Digital revenue by channel
51. Cost per digital transaction
52. Average digital transaction value
53. Digital service uptime records
54. Channel migration records
55. Customer retention records
56. Customer acquisition records
57. Loan processing turnaround time
58. Account opening turnaround time
59. Error or rework records
60. Staff cost by function

1.2 Skills Raw Features

1. Employee training hours
2. AI tool usage frequency
3. Digital skills assessment scores
4. Data literacy assessment scores
5. Cybersecurity assessment scores
6. Reskilling expenditure
7. Human-AI collaboration survey scores
8. Skills demand by role
9. Number of employees trained in digital tools
10. Number of employees trained in AI tools
11. Number of employees trained in cybersecurity
12. Number of employees trained in data analytics
13. Number of employees certified in digital skills
14. Number of employees certified in AI-related skills
15. Training assessment scores
16. Post-training performance scores
17. Employee digital literacy levels
18. Employee AI literacy levels
19. Employee data analytics competency levels
20. Employee cybersecurity awareness levels
21. Employee automation readiness scores
22. Employee innovation readiness scores
23. Employee change readiness scores
24. Employee learning velocity scores
25. Number of staff using AI tools in daily work
26. Number of staff using data analytics tools
27. Number of staff using automation tools
28. Number of staff requiring reskilling
29. Number of staff requiring upskilling
30. Number of roles requiring digital skills
31. Number of roles requiring AI skills
32. Number of roles requiring data skills
33. Number of roles at risk of skill obsolescence
34. Number of roles requiring redeployment
35. Skills gap assessment scores
36. Skills proficiency by department
37. Skills proficiency by job level
38. Digital competency by role

39. AI competency by role
40. Training investment by department
41. Training investment by role
42. Reskilling expenditure by department
43. Reskilling expenditure by role
44. Employee participation in digital transformation projects
45. Employee participation in AI projects
46. Employee participation in automation projects
47. Staff redeployment records
48. Internal mobility records linked to digital roles
49. Digital learning completion rates
50. AI training completion rates
51. Cybersecurity training completion rates
52. Data analytics training completion rates
53. Digital certification pass rates
54. AI certification pass rates
55. Human-centred skills assessment scores
56. Customer advisory skill scores
57. Problem-solving skill scores
58. Change management capability scores
59. Collaboration skill scores
60. Training effectiveness scores

1.3 Employment Raw Features

1. Total headcount
2. Job titles
3. Job descriptions
4. Role categories
5. Number of routine jobs
6. Number of newly created digital roles
7. Number of displaced or automated roles
8. Employee turnover records
9. Number of permanent employees
10. Number of contract employees
11. Number of employees by branch
12. Number of employees by job grade
13. Number of employees by function
14. Number of employees by skill category
15. Number of frontline staff
16. Number of back-office staff
17. Number of operations staff
18. Number of IT staff
19. Number of data and analytics staff
20. Number of digital banking staff
21. Number of customer service staff
22. Number of compliance staff
23. Number of risk management staff
24. Number of credit staff
25. Number of sales and relationship management staff
26. Number of roles affected by automation
27. Number of roles redesigned due to digital transformation
28. Number of roles merged due to automation

29. Number of roles eliminated due to automation
30. Number of roles created due to AI adoption
31. Number of roles requiring AI support
32. Number of roles requiring human–AI collaboration
33. Number of employees redeployed to digital roles
34. Number of employees transferred between departments
35. Number of employees promoted into digital roles
36. Number of employees demoted or downgraded due to role changes
37. Number of vacancies created
38. Number of vacancies filled
39. Time taken to fill vacancies
40. Number of new hires
41. Number of digital-skills-based hires
42. Number of AI-related hires
43. Number of data-related hires
44. Number of employees exiting due to restructuring
45. Number of employees exiting due to automation
46. Number of voluntary resignations
47. Number of retirements
48. Number of retrenchments
49. Number of internal transfers
50. Internal job mobility records
51. Staff absenteeism records
52. Staff productivity records by role
53. Employee workload records
54. Employee overtime records
55. Span of control records
56. Supervisor-to-staff ratio
57. Employee performance rating records
58. Role redundancy records
59. Succession planning records
60. Critical role records
61. Scarce skills records
62. Employee age profile
63. Employee tenure records
64. Employee education level records
65. Employee professional qualification records
66. Employee digital role readiness records
67. Number of hybrid digital roles
68. Number of AI-supervised roles
69. Number of roles with human-centred responsibilities
70. Number of roles requiring continuous reskilling

1.4 Digital Transformation Raw Features

1. Mobile banking usage
2. Internet banking usage
3. Digital onboarding
4. RPA deployed
5. AI-enabled services
6. Chatbot usage records
7. Compliance automation

8. Digital platform adoption records
9. Number of mobile banking users
10. Number of internet banking users
11. Number of USSD banking users
12. Number of agency banking users
13. Number of digital wallet users
14. Number of digital channels available
15. Number of active digital channels
16. Number of digital products launched
17. Number of digitally originated accounts
18. Number of digitally completed account openings
19. Number of digitally abandoned onboarding applications
20. Digital onboarding completion rate
21. Mobile app transaction records
22. Internet banking transaction records
23. Self-service platform usage records
24. ATM self-service usage records
25. Point-of-sale transaction records
26. Card transaction records
27. Digital loan application records
28. Digital loan approval records
29. Digital loan disbursement records
30. Digital loan repayment records
31. Automated fraud detection records
32. Automated AML screening records
33. Automated KYC verification records
34. Number of RPA bots deployed
35. Number of automated workflows
36. Number of digitised manual processes
37. AI-assisted customer service records
38. AI-assisted lending records
39. AI-assisted collections records
40. AI-assisted risk management records
41. AI-assisted compliance records
42. Data analytics platform usage records
43. Business intelligence dashboard usage records
44. Cloud platform usage records
45. API integration records
46. Open banking integration records
47. Core banking system integration records
48. Digital document management records
49. Electronic signature usage records
50. Digital workflow approval records
51. Cybersecurity monitoring records
52. Digital incident records
53. System availability records
54. System downtime records
55. Digital transaction failure records
56. Digital service complaint records
57. Digital customer satisfaction records
58. Digital identity verification usage records

59. Automated customer onboarding records
60. Digital customer journey analytics records

1.5 AI Raw Features

1. Number of AI systems deployed
2. AI use cases by department
3. AI-generated transaction or decision records
4. AI-assisted customer interactions
5. AI decision approval records
6. AI exception or override records
7. AI governance and audit logs
8. Number of AI models in production
9. Number of AI models under development
10. Number of AI pilots or proof-of-concepts
11. Number of AI-enabled processes
12. Number of AI-assisted workflows
13. Number of AI-generated recommendations
14. Number of AI recommendations accepted
15. Number of AI recommendations rejected
16. Number of AI-assisted credit decisions
17. Number of AI-assisted fraud alerts
18. Number of AI-assisted AML alerts
19. Number of AI-assisted KYC decisions
20. Number of AI-assisted compliance reviews
21. Number of AI-assisted risk assessments
22. Number of AI-assisted customer service cases
23. Number of AI-assisted collections cases
24. Number of AI-assisted sales leads
25. Number of AI-assisted marketing campaigns
26. Number of chatbot interactions
27. Number of chatbot-resolved cases
28. Number of chatbot-escalated cases
29. AI response accuracy records
30. AI prediction accuracy records
31. AI false-positive records
32. AI false-negative records
33. AI bias or fairness assessment records
34. AI model drift records
35. AI retraining records
36. AI model validation records
37. AI model approval records
38. AI model monitoring records
39. AI explainability reports
40. AI risk assessment records
41. AI ethics review records
42. AI data privacy review records
43. AI access control records
44. AI user activity logs
45. AI system downtime records
46. AI system failure records
47. AI security incident records
48. AI vendor or third-party system records

49. AI model version records
50. AI training data records
51. AI testing data records
52. AI deployment date records
53. AI cost records
54. AI productivity impact records
55. AI labour substitution records
56. AI labour augmentation records
57. AI-related job redesign records
58. AI-related skills requirement records
59. AI-related staff training records
60. AI-assisted decision share records

2. Engineered Features

These are features created from raw data through indices, ratios, time-based transformations, and interaction terms.

2.1 Index / Composite Features

1. Digital Transformation Index
2. Automation Intensity Index
3. Skill Readiness Index
4. Digital Skills Index
5. Cybersecurity Competency Index
6. Human–AI Collaboration Score
7. Skill Obsolescence Risk Score
8. Skill Mismatch Indicator
9. Employment Elasticity to Digital Transformation Index
10. AI Adoption Index
11. Digital Maturity Index
12. Process Automation Index
13. Digital Channel Adoption Index
14. Customer Digital Engagement Index
15. Data Readiness Index
16. AI Readiness Index
17. Workforce Digital Readiness Index
18. Reskilling Readiness Index
19. Organisational Adaptability Index
20. Digital Productivity Index
21. Labour Productivity Index
22. Workforce Transformation Index
23. Job Automation Exposure Index
24. Routine Task Exposure Index
25. Human Capital Digital Capability Index
26. AI Governance Maturity Index
27. Data Governance Maturity Index
28. Digital Risk Exposure Index
29. Technology Utilisation Index
30. Innovation Capability Index
31. Digital Service Efficiency Index
32. Customer Self-Service Adoption Index
33. Compliance Automation Index

34. AI-Augmented Decision Index
35. Workforce Displacement Risk Index
36. Workforce Augmentation Index
37. Digital Skills Gap Index
38. Future Skills Demand Index
39. Role Transformation Index
40. Organisational Digital Transition Index
41. AI and Digital Transformation Maturity Index
42. Share of AI and DT Index
43. Intelligent Automation Level Index
44. AI-Assisted Decision Share Index
45. Human-Centred Skills Index
46. Employment Structure Index
47. Digital Workforce Exposure Index
48. Role Redesign Intensity Index
49. Automation Substitution Risk Index
50. AI Augmentation Opportunity Index
51. Workforce Reskilling Priority Index
52. Digital Labour Productivity Index
53. Customer Digital Migration Index
54. AI Governance Compliance Index
55. Privacy-Preserving Readiness Index
56. Federated Learning Readiness Index
57. Data Integration Quality Index
58. Model Explainability Readiness Index
59. Regulatory AI Readiness Index
60. Scenario Planning Readiness Index

2.2 Temporal Features

1. Productivity Growth Lag
2. Lagged Productivity Indicators
3. Rolling Averages of Digital Adoption
4. Pre- vs Post-Automation Deltas
5. Average Transaction Processing Time Trend
6. Job Creation Rate Over Time
7. Job Displacement Rate Over Time
8. Employee Turnover Trend
9. Internal Job Mobility Trend
10. Digital Transaction Growth Trend
11. Manual Transaction Decline Trend
12. Mobile Banking Usage Growth Trend
13. Internet Banking Usage Growth Trend
14. Customer Digital Adoption Trend
15. Digital Revenue Growth Trend
16. Operating Cost Trend
17. Cost-to-Income Ratio Trend
18. Revenue per Employee Trend
19. Customers per Staff Member Trend
20. Loan Book Growth Trend
21. Deposit Growth Trend

22. New Account Opening Trend
23. Customer Query Volume Trend
24. Complaint Resolution Trend
25. KYC Processing Trend
26. Compliance Processing Trend
27. Fraud Detection Trend
28. Failed Transaction Trend
29. Reversed Transaction Trend
30. AI Tool Usage Growth Trend
31. AI-Assisted Decision Growth Trend
32. Chatbot Usage Growth Trend
33. RPA Usage Growth Trend
34. Automation Coverage Growth Trend
35. Digital Skills Improvement Trend
36. Training Hours Trend
37. Reskilling Expenditure Trend
38. Skill Readiness Improvement Trend
39. Human–AI Collaboration Trend
40. Routine Job Share Decline Trend
41. Digital Role Creation Trend
42. Role Automation Trend
43. Workforce Redeployment Trend
44. Absenteeism Trend
45. Overtime Trend
46. Staff Productivity Trend
47. Workforce Cost Trend
48. Labour Cost per Employee Trend
49. Branch Transaction Decline Trend
50. Self-Service Usage Growth Trend
51. Time Since Automation Deployment

2.3 Interaction Features

1. Automation Level $\times$ Skill Investment
2. AI Adoption $\times$ Productivity Growth
3. Digital Transformation Index $\times$ Employment Level
4. Automation Coverage $\times$ Error/Rework Rate
5. Training Investment $\times$ Skill Readiness
6. AI Tool Usage $\times$ Human–AI Collaboration
7. Digital Skills Index $\times$ Productivity Growth
8. Routine Job Share $\times$ Automation Intensity
9. Digital Transaction Share $\times$ Frontline Staff Level
10. Reskilling Investment $\times$ Job Mobility
11. AI Adoption $\times$ Labour Cost
12. AI Adoption $\times$ Employment Level
13. AI Adoption $\times$ Skill Obsolescence Risk
14. AI Adoption $\times$ Training Hours
15. AI Adoption $\times$ Job Displacement Rate
16. AI Adoption $\times$ Job Creation Rate
17. Automation Intensity $\times$ Routine Job Share
18. Automation Intensity $\times$ Non-Routine Job Share
19. Automation Intensity $\times$ Labour Cost
20. Automation Intensity $\times$ Revenue per Employee

21. Automation Intensity $\times$ Customer Service Turnaround Time
22. Automation Intensity $\times$ Employee Turnover
23. Digital Transformation Index $\times$ Productivity Growth
24. Digital Transformation Index $\times$ Labour Cost
25. Digital Transformation Index $\times$ Skill Readiness
26. Digital Transformation Index $\times$ Job Creation Rate
27. Digital Transformation Index $\times$ Job Displacement Rate
28. Digital Skills Index $\times$ AI Tool Usage
29. Digital Skills Index $\times$ Human–AI Collaboration
30. Digital Skills Index $\times$ Employee Productivity
31. Digital Skills Index $\times$ Job Mobility
32. Digital Skills Index $\times$ Skill Mismatch
33. Training Hours $\times$ AI Tool Usage
34. Training Hours $\times$ Productivity Growth
35. Training Hours $\times$ Skill Readiness
36. Training Hours $\times$ Human–AI Collaboration
37. Training Hours $\times$ Job Mobility
38. Reskilling Investment $\times$ Skill Obsolescence Risk
39. Reskilling Investment $\times$ Productivity Growth
40. Reskilling Investment $\times$ Workforce Redeployment
41. Reskilling Investment $\times$ Job Displacement Rate
42. Human–AI Collaboration $\times$ Productivity Growth
43. Human–AI Collaboration $\times$ Service Turnaround Time
44. Human–AI Collaboration $\times$ Customer Query Resolution
45. Human–AI Collaboration $\times$ Error/Rework Rate
46. Digital Transaction Share $\times$ Manual Transaction Decline
47. Digital Transaction Share $\times$ Branch Transaction Decline
48. Digital Transaction Share $\times$ Operating Cost
49. Digital Transaction Share $\times$ Revenue per Employee
50. Digital Transaction Share $\times$ Customer Adoption
51. Chatbot Usage $\times$ Customer Query Volume
52. Chatbot Usage $\times$ Call Centre Interactions
53. Chatbot Usage $\times$ Complaint Resolution
54. Chatbot Usage $\times$ Customer Service Turnaround Time
55. RPA Usage $\times$ Back-Office Staff Level
56. RPA Usage $\times$ Processing Time
57. RPA Usage $\times$ Compliance Processing Volume
58. RPA Usage $\times$ Error/Rework Rate
59. AI-Assisted Decisions $\times$ Loan Approval Rate
60. AI-Assisted Decisions $\times$ Fraud Detection Rate
61. AI-Assisted Decisions $\times$ Compliance Processing Rate
62. AI-Assisted Decisions $\times$ Human Override Rate

2.4 Ratio Features

1. Revenue per Employee
2. Cost-to-Income Ratio
3. Digital Transaction Share
4. Automation Coverage Ratio
5. AI-Assisted Decision Rate
6. Customers per Staff Member
7. Digital Transactions per Frontline Staff
8. Revenue per Digitally Trained Employee

9. Reskilling Investment per Staff Member
10. Digital Revenue as a Share of Total Revenue
11. Operating Cost per Employee
12. Income per Employee
13. Transactions per Employee
14. Digital Transactions per Employee
15. Manual Transactions per Employee
16. Transactions per Branch Staff
17. Digital Transactions per Active Customer
18. Mobile Banking Transactions per Digital Customer
19. Internet Banking Transactions per Digital Customer
20. Customer Queries per Staff Member
21. Customer Queries per Call Centre Agent
22. Complaints per 1,000 Customers
23. Complaints Resolved as a Share of Complaints Received
24. Failed Transactions as a Share of Total Transactions
25. Reversed Transactions as a Share of Total Transactions
26. Fraud Cases per 10,000 Transactions
27. KYC Cases Processed per Compliance Staff Member
28. Compliance Cases Processed per Compliance Staff Member
29. Loan Applications per Credit Staff Member
30. Loan Approvals as a Share of Loan Applications
31. Loan Disbursements per Credit Staff Member
32. Loan Book per Employee
33. Loan Book per Credit Staff Member
34. Deposits per Employee
35. Deposits per Branch
36. Active Customers as a Share of Total Customers
37. Active Digital Customers as a Share of Active Customers
38. New Accounts per Onboarding Staff Member
39. Digital Account Openings as a Share of Total New Accounts
40. Branch Transactions as a Share of Total Transactions
41. ATM Transactions as a Share of Total Transactions
42. POS Transactions as a Share of Total Transactions
43. Card Transactions as a Share of Total Transactions
44. Electronic Deposits as a Share of Total Deposits
45. Cash Deposits as a Share of Total Deposits
46. Digital Revenue per Digital Customer
47. Fee and Commission Income per Digital Transaction
48. Non-Interest Income per Customer
49. Operating Cost per Transaction
50. Operating Cost per Digital Transaction
51. Labour Cost per Employee
52. Labour Cost as a Share of Total Operating Cost
53. Training Hours per Employee
54. AI Tool Usage per Employee
55. AI-Assisted Decisions as a Share of Total Decisions
56. AI Overrides as a Share of AI Decisions
57. AI Incidents per AI System Deployed
58. AI Models in Production as a Share of Total AI Models
59. AI Use Cases per Department

60. AI-Resolved Customer Cases as a Share of AI-Assisted Customer Cases

3. Learned Features

These are features automatically learned by the deep learning model after raw and engineered features enter the model.

1. Embeddings
2. LSTM Temporal Representations
3. GRU Temporal Representations
4. Attention-Weighted Feature Representations
5. Multi-Task Shared Representations
6. Scenario-Based Latent Features
7. Learned Workforce Change Patterns
8. Learned Productivity Patterns
9. Learned Skills Transition Patterns
10. Deep Learning Feature Interactions
11. Hidden Labour-Cost Patterns
12. Hidden Employment-Dynamics Patterns
13. Hidden Digital-Transformation Impact Patterns
14. Causal-XAI Feature Attributions
15. Counterfactual Workforce Impact Signals
16. Learned Automation Impact Patterns
17. Learned AI Adoption Impact Patterns
18. Learned Job Displacement Signals
19. Learned Job Creation Signals
20. Learned Role Transformation Patterns
21. Learned Routine-Task Exposure Patterns
22. Learned Non-Routine Work Patterns
23. Learned Workforce Redeployment Patterns
24. Learned Internal Mobility Patterns
25. Learned Productivity-Lag Patterns
26. Learned Digital Adoption Trajectories
27. Learned Skills Obsolescence Patterns
28. Learned Reskilling Impact Signals
29. Learned Human–AI Collaboration Patterns
30. Learned Labour Productivity Representations
31. Learned Cost Efficiency Patterns
32. Learned Digital Revenue Patterns
33. Learned Customer Service Efficiency Patterns
34. Learned Channel Migration Patterns
35. Learned Manual-to-Digital Shift Patterns
36. Learned Compliance Automation Patterns
37. Learned KYC Automation Patterns
38. Learned Fraud Detection Impact Patterns
39. Learned AI Decisioning Patterns
40. Learned AI Override Patterns
41. Learned Model Drift Signals
42. Learned Risk Exposure Patterns
43. Learned Workforce Vulnerability Signals
44. Learned Organisational Adaptability Patterns
45. Learned Scenario-Specific Workforce States

46. Learned High-Adoption Scenario Patterns
47. Learned Moderate-Adoption Scenario Patterns
48. Learned Minimal-Adoption Scenario Patterns
49. Learned Cross-Bank Workforce Patterns
50. Learned Federated Model Representations
51. Learned Department-Level Workforce Patterns
52. Learned Role-Level Transformation Patterns
53. Learned Branch-Level Productivity Patterns
54. Learned Workforce Cost Structures
55. Learned Future Organisation Shape Representations
56. Learned Policy-Relevant Impact Signals
57. Learned Governance and Regulation Signals
58. Learned Explainability-Driven Feature Rankings
59. Learned Counterfactual Role-Change Signals
60. Learned Workforce Transition Pathways